

INNOPOLIS UNIVERSITY · BACHELOR'S THESIS · 2026

Secure Communication for Swarm Robotics

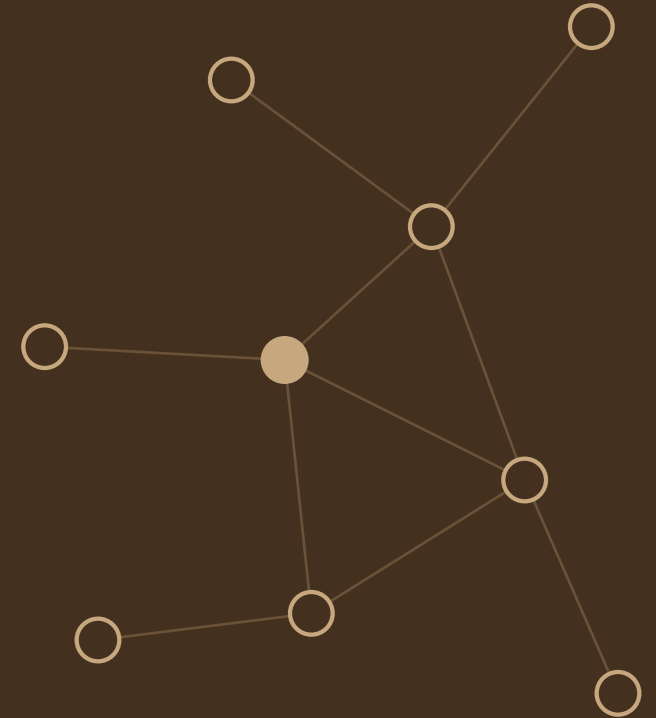
Adaptive lightweight cryptography that switches at runtime.

AUTHOR

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SUPERVISOR

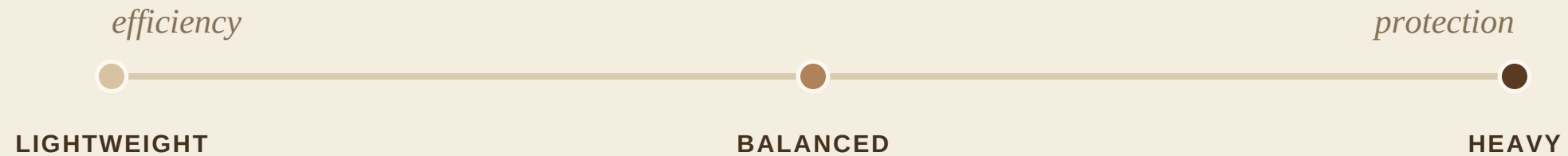
Oleg Bulichev



THE PROBLEM

No single cipher fits a whole mission

Security and performance are coupled —
balance them at runtime, not at design time.



Always heavy

wastes throughput & battery

Always light

unsafe when threat rises

THIS THESIS

Design it formally — then measure it

01

MODEL

Swarm graph + adaptation function

$F : S(t) \rightarrow P$

02

PROTOTYPE

Pairwise testbed, four modes

incl. adaptive switching

03

EVIDENCE

Measured, not just estimated

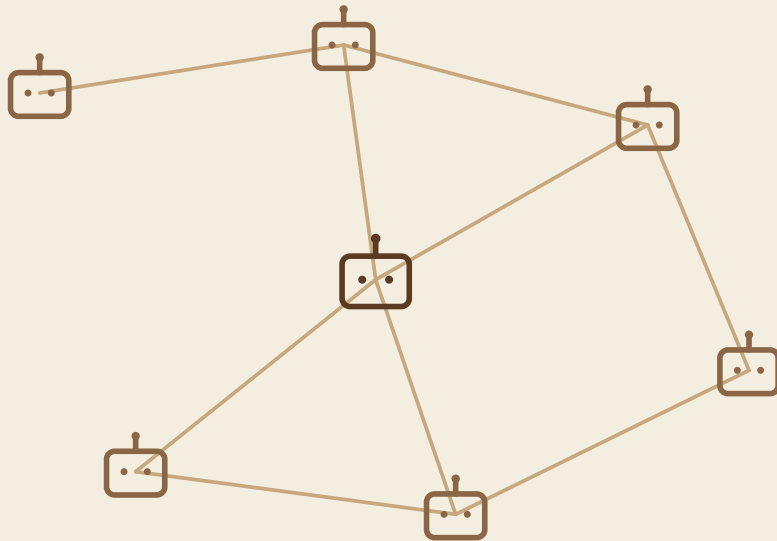
228 benchmark runs

Novelty: a bounded formal model paired with an executable benchmark.

The swarm as a dynamic graph

$$G (V, E, t)$$

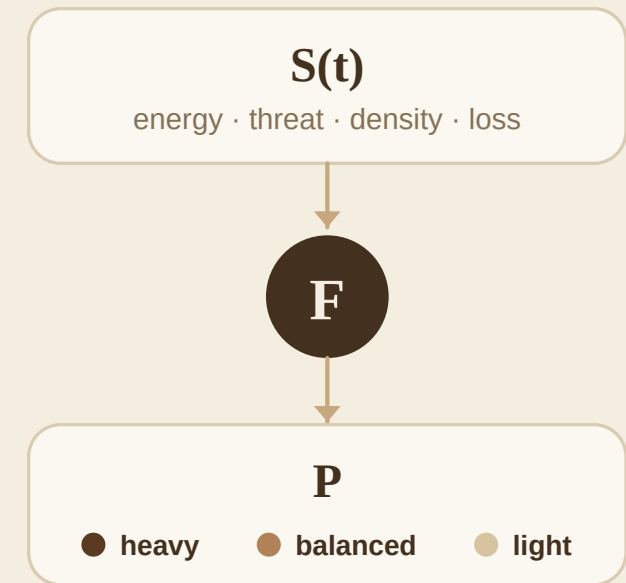
nodes = agents · edges = live links



each node carries energy · keys · position

— THE ADAPTATION FUNCTION

$$F : S(t) \rightarrow P$$



What attackers do — what we guarantee

— THE ADVERSARY CAN



Eavesdrop
read traffic



Tamper
inject / modify



Compromise
leak keys & memory

— THE DATA PLANE GUARANTEES



Confidentiality
AEAD, unique nonce



Authenticity
HMAC / hash token



Integrity & containment
AEAD tag · epoch rekeying

One hybrid stack — each primitive where it fits

SESSION SETUP

X25519 + HKDF-SHA256

ephemeral key exchange per epoch → forward secrecy

periodic

PAYLOAD

AES-GCM / ChaCha20-Poly1305

AEAD confidentiality + integrity

per message

RUNTIME AUTH

HMAC-SHA256 / hash token

lightweight origin & replay check

per message

WHY HYBRID

Costly asymmetric work stays off the per-packet path.

PROFILES

Three operating points

HEAVY

AES-256-GCM

+ HMAC-SHA256

0.250 MiB

crypto work / message

BALANCED

AES-192-GCM

+ HMAC-SHA256

0.250 MiB

crypto work / message

LIGHTWEIGHT

ChaCha20-Poly1305

+ hash token

0.125 MiB

crypto work / message

Lightweight does half the per-message crypto work of the authenticated profiles.

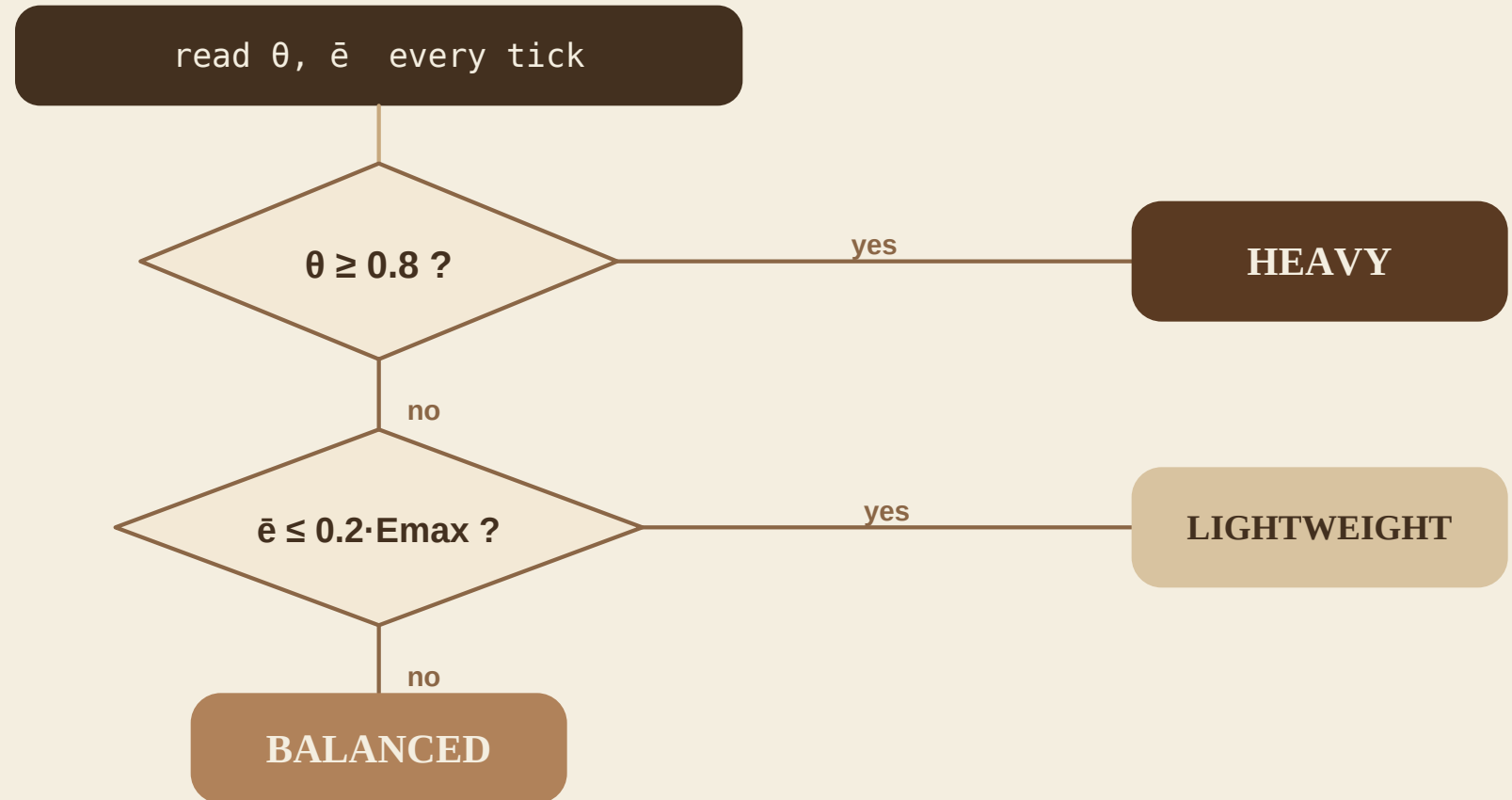
A transparent threshold selector

TWO TRIGGERS

↑ **threat**
escalate to heavy

↓ **energy**
fall back to light

otherwise
stay balanced



BENCHMARK

A constrained, reproducible testbed



two containers · encrypted data path

0.05

CPU CORES

128

MiB RAM

64

KiB / MSG

60

SECONDS

×3

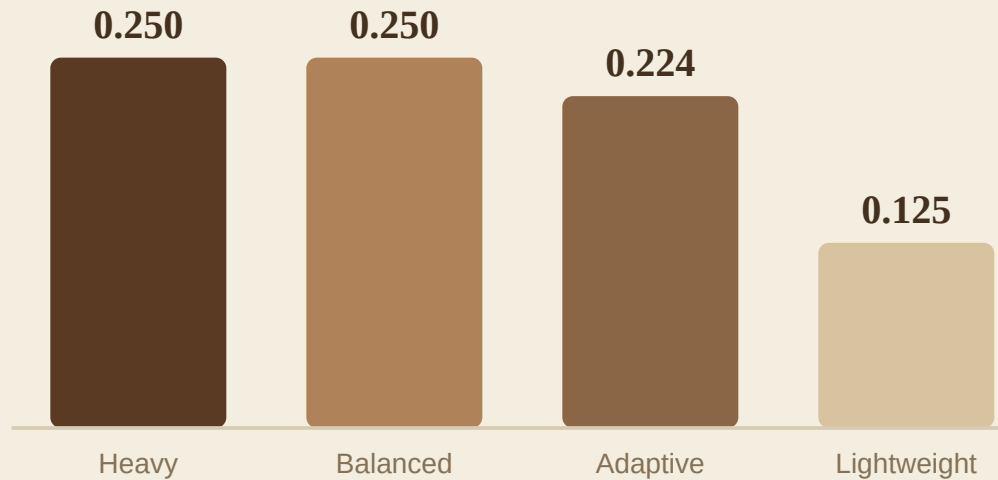
REPEATS

Measured: latency · throughput · drops · switches · crypto work

Adaptive spends protection where it is needed

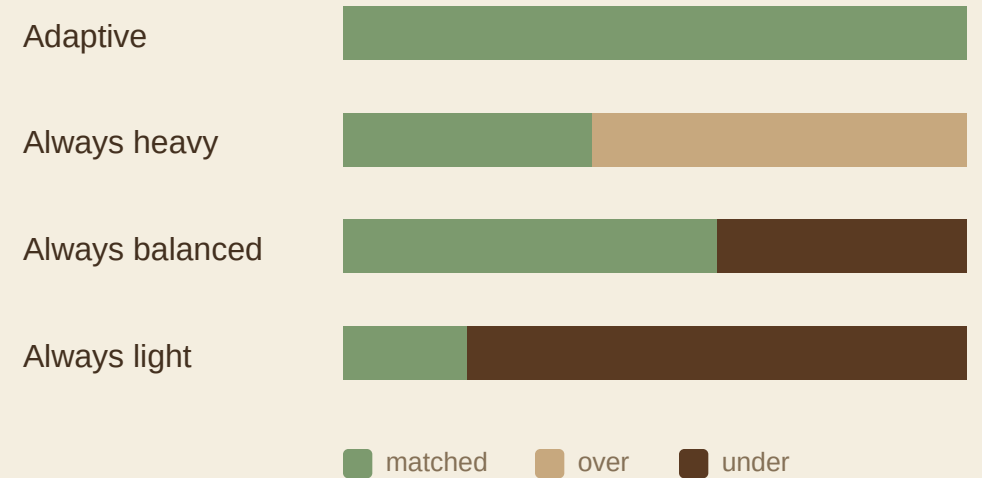
CRYPTO WORK / MESSAGE

MiB · lower is cheaper



POLICY FIT OVER THE 60-s SCHEDULE

share of time at the right protection level

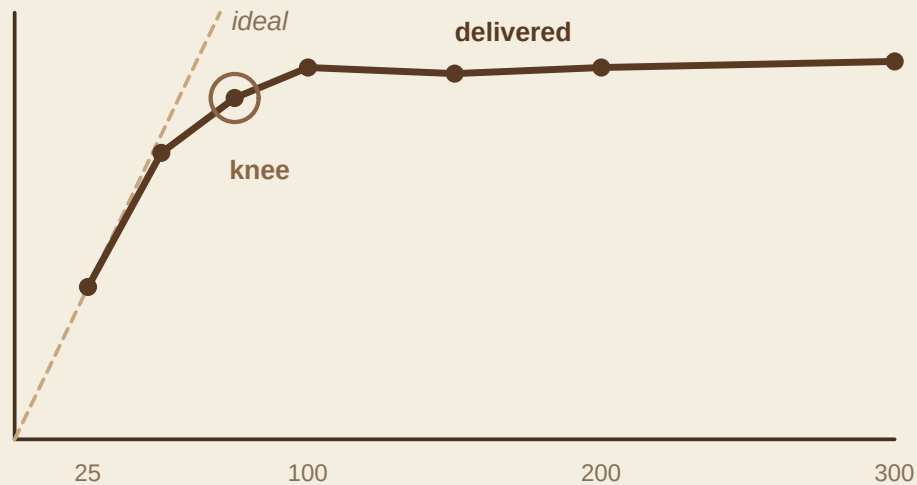


100% of the run at the right level
-26.7% tier-work vs. always-heavy

Under a tight CPU cap, transport dominates

DELIVERED vs OFFERED RATE

0.05-core cap · msg/s



CPU CAP REMOVES THE BOTTLENECK



It's CPU-bound,
not a crypto-design limit.

Near saturation the whole pipeline converges — the profile gap shows in crypto work, not latency.

CONCLUSION

Adaptive crypto — measured, not just proposed

✓ Formalised

swarm graph + $F : S(t) \rightarrow P$

✓ Built

four modes incl. adaptive switching

✓ Measured

lightweight cheapest; adaptive allocates well

WHAT COMES NEXT

01 Authenticated control plane

signed bootstrap & updates

02 Multi-peer validation

churn · contention · async

03 Hardware & post-quantum

real platforms; hybrid PQ

Thank you — questions welcome.

github.com/mazzz3r/adaptive-swarm-crypto-thesis

Limitations & scope

- **Pairwise, not a swarm**

two peers on the measured path; multi-hop & churn are out of scope

- **Energy is a model proxy**

valid for relative comparison only, not battery-accurate

- **Data plane only**

no authenticated control plane (signed bootstrap / updates)

- **Host, not hardware**

Docker + gRPC containers — no embedded radios or microcontrollers

- **Scripted inputs**

threat & energy are scripted, not sensed or estimated

- **Coarse cost metric**

crypto-work counts bytes, so heavy $\approx$ balanced; thresholds set by hand

Every claim is scoped to a constrained pairwise prototype — validated, not extrapolated.